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Digital Television: Impact on Cognitive Development, Educational Inequality and Habits

by

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Abstract

This article assesses the impact of digital television on cognitive development and educational inequality, and studies whether it depends on the educational value of the alternative activities that digital television crowds out. I use heterogeneity in the timing of the television digital switchover in the different postcode areas in England to test for the previous. I find that the digital switchover increased average grades of children by 0.028 standard deviations, contributing to the human capital formation, and that this effect was larger in schools at the bottom of the scores distribution, reducing educational inequality. I also find that the digital switchover decreased the probability of children taking part in harmful activities, and their frequency.

1 Introduction

Since its introduction in 1936, television has been one of the daily habits for most individuals. On average, a person spends more than three hours and a half per day watching TV (Ofcom, 2015). Concerns about its impact on the behaviour emerged rapidly and have been feeded by evidence showing deleterious effects of television viewing on inactivity, aggression, and sleep patterns of children, who are arguably the most sensitive group of the population to televisions influence (Zimmerman, 2005). Cognitive development is not an exception, and it is widely believed by paediatricians that television viewing has a negative impact on school achievement (Gentile et al. 2004).

During the last decade, there has been a positive trend on the average scores of students in standarised exams in England. One example is the Key Stage 2 (KS2) test, an external assessed exam taken by students by the later stage of primary education, of which average grade has increased by 0.78 points. At the same time, it has taken place the introduction of digital television, which has been the first significant revolution in television technology since colour television happened in the 1950s, and has introduced important adjustments in the television market such as an increase in the number of channels from 5 to 40, an improvement in the television definition, additional services related

to multimedia, the possibility of watching television in different languages, and multiplexing, which is the ability of having more than one program in the same channel.

In this paper, I identify whether the digitalization of the television industry had an impact on educational inequality and on the cognitive development of children, measured by their performance at the KS2 test. I also assess whether television exposure impact on education depends critically on the educational value of the alternative activities that it crowds out (Gentkow and Shapiro, 2008), by testing whether digitalization of television changed the extent in which students are involved in harmful activities. Estimating the effect on scores and habits introduces identification problems, as households could self-select into the demand of digital television. To confront potential endogeneity problems, I use a natural experiment consisting on the digital television switchover that took place in UK. Between the end of 2008 and 2012, UK progressively adapted by stages all its TV transmitters, most of which had been built in the 1960s and 1970s, to stop broadcasting analogue TV but strengthen the power of the digital TV signal. At the day when the switchover took place in a transmitter, thousands of individuals obtained access for the first time to an increase in the supply of channels from 5 to 40. The decentralized location of the television transmitters induced substantial geographical variation in the time when the digital switchover occurred across the UK, with neighbouring postcode areas obtaining accessibility to digital television at different years, which created idiosyncratic differences in the receipt of free digital channels. This allows me to use a differences-in-differences approach that compares test scores (habits) across schools (children) and years, attending to the treatment status of each observation. During the switchover process, 10 million of viewers were brought availability of digital signal for the first time, the daily amount of time dedicated to watch television by an average person increased by 23 minutes per day (Ofcom, 2015), and the share of audience of the main five channels decreased by 12% (BARB, 2015).

To conduct the analysis, I use three main datasets. The first was provided by the Department of Education of UK on the aggregated results of pupils at the school level in the KS2 test. The data includes test performance measures of approximately 16,000 schools followed every year since 1991. My second dataset comes from the youth questionnaire in the Understanding Society Survey, where children respond questions about their lives, habits, or social activities they take part in. This allows me to test the hypothesis of whether television exposure impact on education depends critically on the educational value of the alternative activities that it crowds out. Finally, I obtain my third dataset by webscraping data, on the DigitalUK webpage, of the date when the digital switchover took place at each postcode in England. After having assembled my third dataset, I match it with the two former ones.

A key finding is that the digital switchover increased the average KS2 test scores by 0.028 standard deviations. The impact was higher in schools with a higher proportion of students who were economically disadvantaged or who had other language than English as a first language. Importantly, the magnitude of

the impact depends on the academic performance previous to the switchover, with a stronger effect on schools below the median, and no impact on schools above it. This points that the digital switchover reduced educational inequality, which suggests that television exposure impact on education depends critically on the educational value of the alternative activities that it crowds out (Gentkow and Shapiro, 2008). I find evidence supporting this hypothesis. My preferred estimates indicate that digital switchover reduced the probability of a kid having ever drunk alcohol by 7.31 percentage points, and also decreased the frequency in which kids consume alcohol. There are two possible mechanisms driving the results in scores and habits: either digital television changed the time allocation of low performance students, who substituted TV watching time for detrimental activities, or it increased the educational value of TV contents. I name the former as the quantity mechanism, and the latter as the quality mechanism. I find that the digital switchover increased happiness in school and motivation to get further education, but did not alter time allocation.

The present analysis is related to empirical studies that study the impact of television on the cognitive skills development and reading habits formation (Beentjes et al. (1988); Walberg & Tsai (1985); Rice (1987)). Whereas most of the literature used to highlight a negative relationship between TV and cognitive development (Christakis et al. (2004); Timmer et al. (1985); Hancox et al. (2005)), recent research has shed light on a potential positive impact (Gentkow & Shapiro (2008)). This paper is also related to empirical papers on the importance of TV contents on behavior and academic skills accumulation (Wright et al. (2001)). There is evidence of academic achievement being lowered when exposed to pure entertainment and violent content (Kirkorian, Wartella, and R. Anderson, 2008), adult aggression increasing due to TV-violence viewing (Huesmann, Moise-Titus, Podolski and D. Eron, 2003) and harmful habits becoming more frequent because of TV watching (Robinson, L. Chen and D. Killen, 1998), but also there is evidence of cognitive and academic enhancement as a response of exposure to age appropriate programs designed around an educational curriculum (Bryant et al. (2004); Fisch (2004); Calvert (2003)). This ambiguous evidence may be explained by TV programs displacing differently non-TV activities depending on their content (C. Huston et al (1999)). From a policy perspective, if TV contents alter cognitive development, deregulation of content availability may bring a considerable effect on educational attainment. Last but not least, my paper relates to the empirical literature on the importance of the TV industry on social attitudes, perceptions and behaviour (Jensen and Oster (2009); La Ferrara (2007); Mastrorocco and Minale (2015); Gentkow (2006); DellaVigna and Kaplan (2007)).

This paper contributes to the previous literature in that (i) I examine the access to digital television, which caused a change both in the time that people view TV and in the contents they watch, instead of television, (ii) I use a large panel dataset based on external assessed examinations to study the impact on cognitive development, instead of using survey or school self-reported data (iii) I use a difference in difference approach at the finer geographical level of the postcode area, and (iv) I use an additional panel dataset based on the youth

questionnaire of the Understanding Society Survey, to test whether the impact on test scores comes from a change in habits, and to study the mechanism that drives the results in scores and habits.

The remainder of the paper is organized as follows. Section II describes digital TV industry and its introduction. Section III presents the data. Section IV contains the empirical strategy and results on education. In Section V presents estimates of the effect of digital television on habits, and of the mechanism that drives the results. Section VII concludes.

2 Digital Television

Digital television has been the first significant revolution in television technology after colour television happened in the 1950s. It has several advantages over its analogue predecessor, such as the provision of more digital channels in the same space, higher definition, additional services related to multimedia, the possibility of watching TV in different languages, and multiplexing, which is the ability of having more than one program in the same channel. However, the most important feature of digital television is the increase from the 5 channels that could be watched through the analogue aerial to the 40 different channels that can currently be watched through the digital signal.

In 2004, The British Government delegated Ofcom.[1] and Digital UK.[2] jointly the task of discontinuing analogue television broadcasting on a transmitter group.[3] basis. Meanwhile Ofcom amended the broadcast licences requiring the delivery of the analogue switch-off according to a specific timetable, Digital UK organised the programme of the digital switchover, managed the upgrade of the television transmitters, and ensured that individuals were supported and understood the process. The digital switchover, or cessation of the analogue television transmission, was done in stages, from the end of 2008 to 2012, where transmitter groups ceased broadcasting analogue television services at different deadlines to start transmitting high-power digital signal. The switchover process is approximately represented in Figure 1. More accurate details of the switchover timetable can be found in Table A.1 of Appendix.

[1] Ofcom is the communications regulator in the UK

[2] Digital UK is the independent, not-for-profit organisation, funded by the public-service broadcasters and multiplex operators

[3] A transmitter group is a bunch of transmitters, one principal transmitter and several relay transmitters, each of which provides TV signal. The former broadcast TV signal to individuals and to a number of the later type of transmitters, which repeat the TV signal to many other individuals who could not receive TV signal from former.

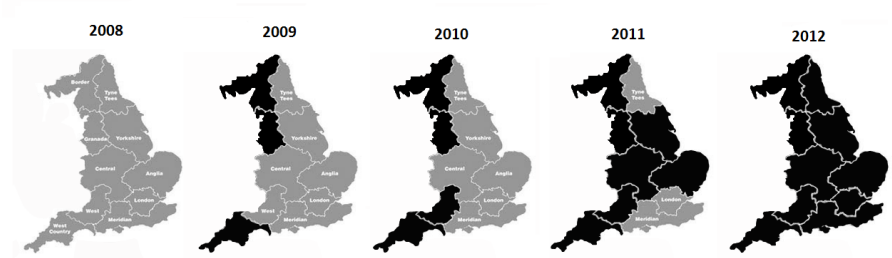

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Figure 1: Stages of the Digital Switchover in England

Digital signal in the television market in UK is provided by 1150 separate transmitters serving to 26 million of homes. There are 81 main transmitters located across different parts of the UK, and 1,154 relay transmitter sites, which receive TV signal from the former and repeat it to those households living far from the main transmitters. Before switchover, one-in-four households could not receive digital television via an aerial. This was independent of whether these households had already adapted their television set in order to be able to receive digital TV. The reason is that, TV transmitters were broadcasting digital and analogue signal at the same time, which decreases the strength of the coverage of the prior signal. As a consequence, digital signal could only reach households who were both located close to a TV transmitter and not affected by adverse geographical factors. At the time when the switchover happened in a transmitter group, it ceased broadcasting analogue television services and started transmitting high-power digital signal, making digital signal available for the latter group of households.

Importantly, each postcode area in England receives TV signal from several and different TV transmitters, each of which may have different digital switchover dates. This is the reason why I define the date when the digital switchover occurred in a particular postcode area as the earliest date among the switchover dates of the TV transmitters providing signal to that postcode. The switchover process made digital TV through an aerial (Freeview) available to all homes, with 10 million of viewers been brought into Freeview coverage for the first time. Additionally, there were millions of households who could receive digital signal through an aerial before the switchover, but who had not still adapted their TV sets to receive it. In fact, only three years before the switchover started, more than three-quarters of homes were still analogue on at least one TV (Ofcom 2012). As a consequence, there were additional millions of short-sighted individuals who were able to watch digital TV only weeks or days before the switchover happened.

Digital switchover may have had an impact on the cognitive development of children through a quantity or a quality mechanism. The former means that the higher variety in the TV programs induced children stop doing alternative activities to watch more TV. The latter means that the increase in the number of channels brought a change in the TV contents watched by children, and so

in the intrinsic values that children obtain from watching TV.

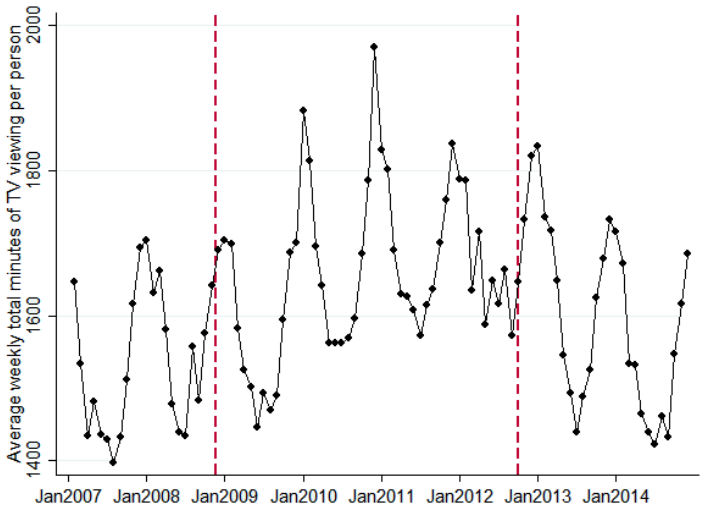

Figure 2: Quantity channel

The quantity mechanism came in the form of an increase in the time dedicated to watch TV by children, from 134 minutes per day before switchover to 142 by the end of it (Ofcom, 2015). After 2012, year in which the switchover finished, there has been a continuous decrease in the average time of TV watching in UK. Possible explanations are the growing popularity in take-up of non-broadcast, on-demand services such as Netflix and Amazon Prime Instant Video, as well as the growth in connectivity and mainstream adoption of smartphones and tablets that have taken place since 2012, and that are likely to have reduced traditional and broadcast TV-set viewing (Ofcom, 2015). Figure 2 represents the quantity mechanism.

The quality mechanism came from a drastic decrease in the audience share of the main five channels after the switchover , from 63.5% in 2007 to 51% in 2013, in favour of the newly available digital channels (Ofcom, 2015). The new channels have different TV contents than the main channels, and so transmit different intrinsic values, which could have affected behaviour and cognitive development of children. Digital television also allowed viewers with the possibility of watching TV in different languages, which may have stimulated the verbal abilities of kids. Figure 3 represents the quality mechanism.

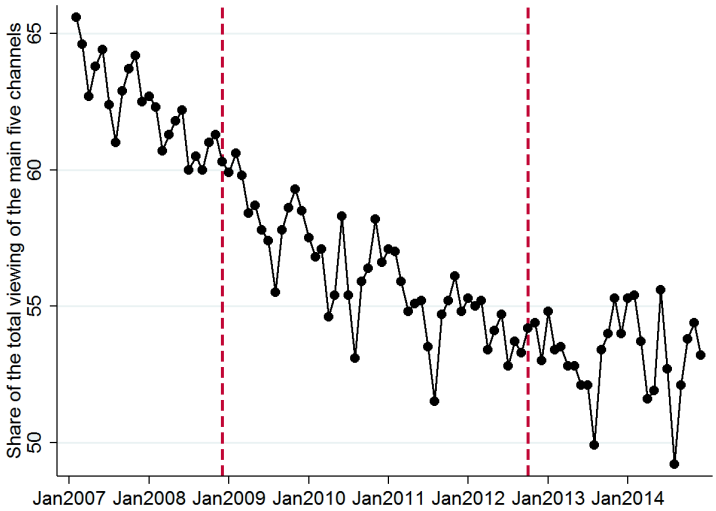

Figure 3: Quality channel

3 Data

3.1 Schooling Data

I obtained my data on academic performance from the Department of Education of UK. This dataset contains academic information of students in the KS2 exam, aggregated at the school level. In particular, the dataset is a panel that includes KS2 test performance measures of 16,000 schools in England, followed every year during the period 1991-2016. KS2 test is an external assessed exam taken by eleven years old students by their later stage of primary education in the month of May, and consist of sections on mathematics; English grammar; punctuation and spelling, and writing. KS2 test is graded in a scale of 0 to 33 points, even though most of the schools are centered around the national average point score, which is 27. Figure 4 represents the average grade distribution of schools during the different years that I use in the analysis. A mapping of point scores to levels can be found in the Appendix A of the paper.

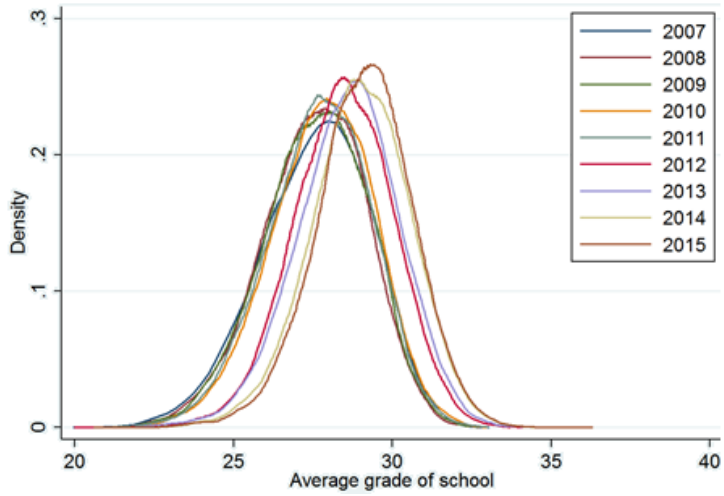

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Figure 4: Yearly average grades distribution

There have been two important reforms in the structure of KS2 exams. Firstly, this test was based in an English, Maths and Science part until 2009. However, the latter ended in 2009, and Average Points Score were recalculated by the Department of Education back to 2007 on the basis of English and Maths test results, making 2007-2009 comparable with later years. Because of the change in the exam format, I only include the years after 2006 in the analysis. Secondly, before 2012, the maximum grade that could be obtained in the KS2 exam was 35, but from 2012 onwards, the highest score attainable increased to 39. In addition, since 2012, the results in the writing part of the KS2 exams have been based on the teacher assessment of the work that students have made during the last year of primary education, instead of being externally assessed. During the analysis, I use time fixed effects and school individual trends to control for these changes.

The dataset has information on several academic performance indicators for each school, such as the average test score, or the percentage of students with a level of 4 or higher in the Maths and English section, or in the overall test, among others. The 'Key Stage 2 average point score' is a much more granular measure of attainment than the 'Percentage of pupils achieving the national expected level or higher', because it shows how far are students performing with respect to the national expected level. In addition to these academic performance measures, there is information on school characteristics such as the total number of students, or the percentage of students with a language other than English as the first language, among others.

The schools in the dataset are state-funded schools. I exclude Community Special Schools and Foundation Special Schools from the analysis since these type of schools have only been operating since 2010. In addition, as independent schools are not required to enter their pupils for the tests, the Department

of Education does not publish KS2 results data for them. Neither there is information of schools with cohorts of fewer than 6 pupils, as the Department of Education does not publish their results due to its privacy policy. Within schools, every student in year 6 of primary education is eligible to take the key stage 2 test. An eligible student may still not take the test if they have not yet reached the level of ability required to sit these tests, or if a disability prevents them from being able to access the tests. However, the Department of Education includes every child registered for tests in their calculation of performance measures, regardless of whether they actually sit the test, and only accepts a school's request to remove a child from the calculations if the student has a limited English and has arrived from a non-English speaking country within the last two years.

3.2 Digital television data

I obtained my data on the date when the digital switchover took place at the postcode area level by webscraping the DigitalUK organisations webpage, which provides information about which TV transmitters that provide signal to a specific postcode area, and about the quality of the signal. More generally, this dataset contains information on the year when the digital switchover took place on each of the postcodes contained in my schooling dataset. The importance of having postcode level data lies in the fact that each postcode area receives TV signal from several TV transmitters, which may have had different digital switchover dates. As a result, I define the date of the digital switchover in a particular postcode area as the earliest date among the switchover dates of the TV transmitters providing signal to that postcode. Figure 7 draws a hypothetical postcode area receiving digital signal from different TV transmitters, located at different TV regions. These are: the main transmitters A, B, C, E, and the relay transmitter that receives signal from main transmitter D. Assuming that the main transmitters A-E have different digital switchover dates, which is the same than the date of their respective relay transmitters, then the digital switchover date for the green postcode area is the earliest among the switchover dates of transmitters A-E.

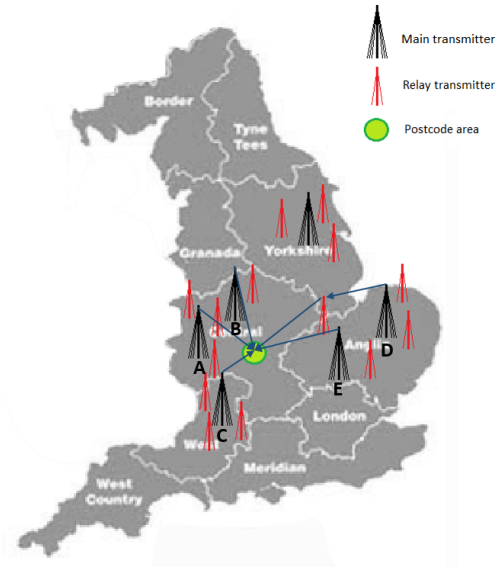

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Figure 5: Explanation of digital switchover

The implicit assumption I make to match my educational and digital TV datasets is that students live in the postcode area of the school they attend, and so, students of a school are affected by the digital switchover at the time that it occurs in the postcode area of the school. This is consistent for two main reasons. Firstly, most of schools in UK receive more applications than places available, and distance from the permanent home address of the child to the school is one of the main factors taken into consideration when selecting which students to admit. Secondly, I do not expect children to attend a school far away from home, given the time costs that would be incurred. The match between the Department of Education data and the digital TV data reduces the initial sample of schools to 160,776 observations.

3.3 Understanding Society survey data

My data on time use and leisure comes from the youth questionnaire of the Understanding Society Survey, with geographical special license at the Lower Layer Superoutput area (LSOA).[4] unit. The Understanding Society survey is a UK Household longitudinal survey of the members of approximately 40,000 households in the United Kingdom during the period that goes from 2009 to 2014. Households recruited at the first round of data collection are visited each

[4]LSOAs are geographical units designed with a minimum population figure of 1000 and a mean population figure of 1500 and each consists of a grouping of Output Areas, typically five

year to collect information on changes to their household and individual circumstances. As part of the Understanding Society, approximately 3,000 household members aged from 10 to 15 years old are asked to complete a self-completion youth questionnaire about their lives, experiences, habits, or the social activities in which they get involved. In particular, the youth survey contains questions on how happy they are with their school and own work; on the leisure activities they are involved in, such as the frequency in which they watch television, use computer, play with game console, go out during the night, smoke, or drink alcohol, among others; and on demographic characteristics such as age, sex and parents income. The analysis of the impact that the digital switchover has on some of these leisure activities is important to test the hypothesis of whether television exposure impact on education depends critically on the educational value of the alternative activities that it crowds out.

The Understanding society survey contains an individual identifier variable unique to each sampled child answering to the youth questionnaire, which allows me to use child fixed effects. I use the LSOA, which is the smallest geographical unit that I can observe in the survey, to include information about the date when the switchover takes place in the postcode area where each individual interviewed lives. There are 34,378 LSOAs in England and Wales, and I randomly select a postcode within each of the LSOAs where the individuals interviewed live. This does not constitute a problem for the analysis, given that the LSOA is a very precise geographical unit, so that when the switchover happens in any of the postcodes of a particular LSOA, it is very likely that it has occurred at the same time for the entire LSOA. I then use Java software to webscrape information about the date when the digital switchover takes place on each of the postcodes where individuals interviewed live, and combine this information with the Understanding Society dataset using the postcode as a merging unit, which reduces the initial sample of individuals to 15,069 observations.

4 Empirical Framework

4.1 Summary Statistics

The digital switchover was implemented in stages across UK, so that schools in England, and their respective postcode areas, were affected by it at different points in time. Figure 6 represents the percentage of schools that had already been affected by the switchover on May, which is the month when students do the KS2 exam, of every year during the period 2007-2014. It is important to study whether schools affected at different points in time by the digital switchover were already different in my variables of interest before the process had even started. Column 1 of Table 1 presents unweighted summary statistics on the 16,000 schools in my sample. In the year before the switchover process started, the average grade of a school was 27.55 points, and 27.58 only

two years before. I also include summary statistics of the academic performance of schools after the switchover process had started, and of demographic characteristics such as the number of pupils studying in a school, the number of students with special needs, the percentage of students with English as first language, the percentage eligible for free school meals, and religion status of the educational institution.

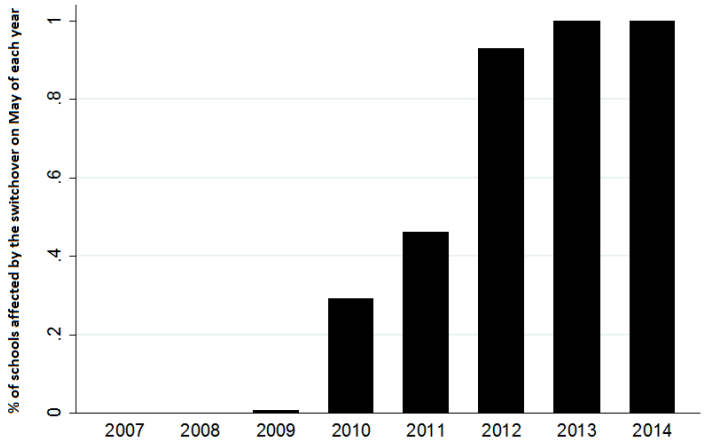

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Figure 6: Percentage of the territory affected by the Digital Switchover

Columns 2 - 5 compare four different groups of schools, assigned using the year when the school was affected by the digital switchover. In particular, my first group of schools is formed by those affected by the digital switchover in 2009, the second group by those affected in 2010, and so on. On the one hand, educational institutions with late access to digital television have a higher number of students and pupils with special needs, suggesting that urban areas were affected by the switchover process at a later stage. On the other hand, early adopters seem to have a higher proportion of students with English as a first language and a lower percentage of students eligible for free school meals, which suggests that early adopters are socioeconomically advantaged. Importantly, all the groups seem to have similar average grades in 2007 and 2008, years before the switchover started, but also none of the groups seems to follow a different pattern in tests performance once the digital switchover has been introduced respectively on each of them. This advocates for a null effect of the digital switchover on grades. The descriptive statistics, however, do not it take into account potential differences across schools and time, possible dissimilarities in performance trends, demographic composition, nor weights the schools depending on their total number of students. In my differences-in-differences approach, I estimate the impact of the digital switchover on grades taking into account

these factors.

Table 1: Descriptive statistics

	Total	2009	2010	2011	2012
N of students	260.95	227.31	221.97	255.26	308.50
	147.50	123.40	132.80	145.39	160.85
N of students with SEN	3.12	2.64	2.36	3.07	3.78
	2.30	1.92	2.02	2.29	2.50
% of students free meals	0.27	0.27	0.20	0.25	0.30
	0.19	0.19	0.16	0.18	0.20
% students English 1st language	0.87	0.92	0.94	0.89	0.77
	0.22	0.17	0.10	0.20	0.26
Average grade 07	27.58	27.63	27.74	27.52	27.61
	1.75	1.71	1.71	1.76	1.77
Average grade 08	27.55	27.62	27.66	27.48	27.56
	1.65	1.61	1.70	1.68	1.62
Average grade 09	27.58	27.65	27.66	27.50	27.64
	1.66	1.61	1.63	1.69	1.66
Average grade 10	27.78	27.88	27.84	27.66	27.89
	1.62	1.57	1.62	1.66	1.59
Average grade 11	27.69	27.77	27.83	27.57	27.80
	1.64	1.62	1.64	1.67	1.58
Average grade 12	28.44	28.48	28.52	28.33	28.58
	1.58	1.55	1.57	1.61	1.53
Average grade 13	28.60	28.62	28.69	28.47	28.80
	1.62	1.57	1.66	1.68	1.55
N	14603	3487	756	6498	3862

4.2 Possible selection problems

In this paper, I compare the academic performance of different treatment groups of schools that were affected by the digital switchover at different times during the period 2008-2012. One possible threat to the analysis is that the assignment of treatment is not random. For example, digital switchover could have been implemented earlier in areas where there is a higher demand of TV time viewing, which may be correlated with the academic performance of children. Therefore, it is important to study the nature of the digital television switchover, to check whether digital television expanded first in regions that were performing differently in KS2 tests than later adopters. If this is the case, the effect that the digital switchover has on grades would be only capturing previous grading trends.

To face possible selection problems, I estimate whether previous grade trends or demographic characteristics of schools predict the timing of the digital switchover. To state this approach more formally, let y_s be a variable that takes value 1 if the school s is a late adopter of digital television (if it is affected by the digital switchover in 2011 or after) and 0 if the school is an early adopter

(when it is affected before 2011). Let $ag_{s,07}$ and $ag_{s,08}$ be the average grade of school s in the years 2007 and 2008 respectively. $X_{s,07}$ and $X_{s,08}$ are a set of controls that include the total number of students with special needs and the total number of students in school s in 2007 and 2008 respectively. Therefore, I estimate the following probability model:

$$y_s = \alpha + \beta ag_{s,07} + \lambda ag_{s,08} + \phi X_{s,07} + \gamma X_{s,08} + \varepsilon_s \tag{1}$$

Where ε_s is a school level error term. A first specification (column 1 of Table 2) addresses the concern of whether previous average grades have any effect on the date when the digital switchover takes place for the different postcode areas, finding that higher grades in the year before the start of the switchover process decreases the probability of being a late adopter. After controlling for demographic controls for the years before the switchover (column 2 of Table 2), I find that the previous effect becomes insignificant but that higher grades two years before the switchover increases the probability of being a late adopter. A late adoption of the digital switchover is also more likely in larger schools: a one percent increase in the size of a school in 2008 is associated with a 0.2 percentage point increase in the likelihood of a school being affected by the digital switchover in a late stage. Given that bigger schools tend to be in urban areas, this result suggests that digital switchover was implemented at first place in a higher proportion of rural areas, leaving cities for the end of the process. Schools with a higher number of students with special needs in 2007 and 2008 are also more likely to be late adopters.

In column 3, I add local educational authority fixed effects to control for geographic heterogeneity in the digital switchover penetration, capturing the determinants of within-local educational authority digital TV availability. Neither previous average grade trends nor past values of schools demographic characteristics have an impact on the implementation of the digital switchover in the different areas. In particular, previous grades seem to be negatively correlated with the probability of a school being a late adopter, although insignificant.

In sum, I find that previous grades trends are uncorrelated with the timing at which the digital switchover takes place in the different postcode areas and that previous trends of schools demographic characteristics are also orthogonal to the timing of the digital switchover, once I have controlled for local education authority fixed effects.

4.3 Impact of Digital Television on School Achievement

The key advantage of this paper is to identify the effect of the digital switchover on KS2 cognitive development measured in KS2 test scores, by exploiting postcode areas specific idiosyncratic deadlines in the timing in which TV transmitters were upgraded. This allows me to implement a difference-in-difference design that compares the academic performance of schools affected by the digital switchover at different dates, and before and after the switch

Table 2: Exogeneity

	Late introducer	Late introducer	Late introducer
Average Grade 07	-0.00203	0.00580*	-0.000722
	(-0.62)	(1.68)	(-0.56)
Average Grade 08	-0.00641*	-0.00264	-0.000175
	(-1.84)	(-0.72)	(-0.13)
N students SEN 07		0.00714***	-0.000362
		(4.20)	(-0.57)
N students SEN 08		0.00300*	0.000502
		(1.85)	(0.83)
log N students 07		-0.103	-0.0201
		(-1.40)	(-0.74)
log N students 08		0.199**	0.0281
		(2.74)	(1.04)
LEA fixed effects?	NO	NO	YES
Constant	0.947***	0.502***	1.020***
	(13.44)	(6.34)	(6.21)
Observations	12822	12358	12358

t statistics in parentheses
* p<0.10, ** p<0.05, *** p<0.001

to digital signal occurred. In addition, I address the hypothesis that television exposure impact on education depends critically on the educational value of the alternative activities that it crowds out, by estimating my difference-in-difference approach in the quartiles subsamples of the grades distribution in the year before the switchover started . To state this approach more formally, let $y_{s,t}$ be the average grade of students in school s in academic year t. Given the geographic information at the postcode level in my digital TV dataset, I am able to distinguish when each school was affected by the digital switchover. Let $DigitalSwitch_{s,t}$ be a dummy that takes value 1 if school s has already been affected by the digital switchover at year t, and 0 otherwise. Therefore, my baseline specification is:

$$y_{s,t} = \alpha + \beta DigitalSwitch_{s,t} + \gamma X_{s,t} + \delta_s + \lambda_t + \delta_s t + \varepsilon_{s,t} \tag{2}$$

Where δ_s are school fixed effects meant to capture any unobserved time-invariant characteristics that affect academic performance and which may be correlated with the timing of the switchover to digital TV. λ_t are time year fixed effects to allow for flexible trends in grades common across schools. $\delta_s t$ are school individual trends, and controls for possible different trends in grades across schools. $X_{s,t}$ is a set of demographic school characteristics, which change across schools and over time, such as the total number of students or the num-

ber of students with special needs. As discussed in the exogeneity analysis, school demographic characteristics seem to be orthogonal to the timing of digital television switchover once I have controlled for geographical fixed effects. However, given that they are observable predictors in the summary statistics, and in specification (1) when I do not control for geographical fixed effects, it is worth to include them as controls, in order to limit the chance of my results being confounded by unobserved differences in time trends across schools of different size or with a different proportion of students with special needs. $\varepsilon_{s,t}$ is a school and time specific level error term. Throughout the empirical analysis, I cluster standard errors at the school level to allow for an arbitrary correlation of residuals within schools. My identifying assumption is that, conditional on school and year fixed effects, school individual trends, and demographic time-varying controls, the timing of the switchover process is idiosyncratic.

I firstly estimate my baseline specification without school individual trends and demographic controls (Column 1 in Table 3). Digital switchover increases averages grades of schools by 0.039 standard deviations, which means that overall, students affected by digital switchover perform better. In column 2, I add school demographic controls, which decreases the magnitude of the impact, but not its sign neither its significance. As expected, the average grades are positively correlated with the size of schools but negatively correlated with the number of students with special needs in a school. In column 3, I estimate my baseline specification including school individual trends, to control for potential differences in trends of average scores among schools. Once I control for this, Digital switchover increases averages grades of schools by 0.028 standard deviations.

In my previous benchmark specification (column 3), I only consider the effect of the introduction of digital television, without studying how this impact evolves over time. To account for this, I control for the variable $DigitalTenure_{s,t}$, which is equal to the number of years that has occurred since the date of the switchover, in the postcode area of school s at time t. In particular, variable $DigitalTenure_{s,t}$ takes value 1 if the switchover has occurred within the last 12 months at time t, value 2 if the switchover occurred earlier than 12 months ago but later than 24 months ago at time t, and so on. Formally:

$$y_{s,t} = \alpha + \beta DigitalSwitch_{s,t} + \phi DigitalTenure_{s,t} + \gamma X_{s,t} + \delta_s + \lambda_t + \delta_s t + \varepsilon_{s,t} \quad (3)$$

In column 4 of Table 3, I estimate (2) including school demographic characteristics, school fixed effects, time fixed effects and school individual trends. The estimate of the effect of Digital switchover on school achievement is positive (0.034 standard deviations), significant, and higher than in previous specifications. In contrast, an additional year since digital switchover occurred decreases the average grade of a school by 0.01 standard deviations, which suggests that the positive effect that the accessibility to digital television has on grades disappears after two years.

In columns 5-8, I assign schools into quartiles attending to the average

grade distribution in 2008 (before the switchover started), and then implement (2) on each of these quartile groups, to study the hypothesis of whether television exposure impact on education depends critically on the educational value of the alternative activities that it crowds out. According to this hypothesis, digital television availability crowds out lower educational activities for low performing students, improving their attainment, but also crowds out higher educational value activities for the high attainment ones, deteriorating their achievement. I find that schools at the lowest and at the second lowest quartile of the distribution perform better as a response to the digital switchover, with an increase of 0.071 and 0.097 standard deviations respectively, which is higher than the overall increase of 0.034 standard deviations. In contrast, digital switchover does not have an effect on the top quartiles of the distribution, suggesting that the positive impact that the digital switchover has on grades was driven by the schools that were performing worse previous to the digital switchover started. This gives partial evidence for the hypothesis that television exposure impact on education depends critically on the educational value of the alternative activities that it crowds out. Low performance students may be substituting television time viewing for harmful activities (quantity mechanism), or new TV contents may be motivating them to have better habits and interest in obtaining more education (quality mechanism). In contrast, digital switchover does help high performance students, suggesting that neither high achievers change their time allocation, nor new TV contents affect their behaviour or interest towards schooling. Importantly, the effect of an additional year of accessibility to digital TV decreases average grade by 0.017 and 0.060 standard deviations for the lowest and the second lowest quartile respectively, so that the positive effect of the digital switchover disappears after two years. An additional year after the switchover has no impact on the grades of the schools in the top quartiles, given the lack of effect of the digital switchover on average scores for these groups. It is relevant to study the timing of the switch-over in a more profound manner.

4.4 Timing of the Digital Switchover

My benchmark specification does not give evidence of the dynamics of the impact of digital television accessibility on educational achievement: whether the effect accelerates, stabilizes or even mean reverts. There may be previous trends in grades as the number of households watching digital television were already increasing before the switchover happened. However, if it is the case that previous trends in grades or some other factor related to them led the adoption of the digital switchover, then the previous estimates would obscure this reverse causality. Exploiting the fact that different postcode areas were affected by the digital switchover at different points in time between 2008 and 2012, I explore these dynamics in Table 2 , which provides estimates of a set of models that incorporate a different number of leads and lags of the digital switchover. Column X presents my benchmark specification, where I add indicator variables from the second year prior to the third year after the switch-off of the analogue TV

Table 3: Effect of digital switchover on KS2 Test Scores

	Standarized KS2	Standarized KS2	Standarized KS2	Standarized KS2	Quartiles Standarized KS2 Test Average Grade			
	Test School Average Grade	Test School Average Grade	Test School Average Grade	Test School Average Grade	Q1	Q2	Q3	Q4
Digital Switchover	0.0394***	0.0303***	0.0281**	0.0340**	0.0713**	0.0974***	-0.0130	-0.0241
	(4.49)	(3.53)	(3.09)	(3.09)	(2.86)	(4.46)	(-0.60)	(-1.19)
Digital TV tenure				-0.0104	-0.0173	-0.0595**	0.0300	0.0145
				(-0.89)	(-0.66)	(-2.48)	(1.27)	(0.67)
N students with SEN		-0.0633***	-0.0640***	-0.0641***	-0.0584***	-0.0609***	-0.0671***	-0.0732***
		(-59.08)	(-55.98)	(-56.03)	(-29.61)	(-30.02)	(-27.50)	(-21.89)
log of N of students		0.306***	-0.427***	-0.427***	-0.368***	-0.213**	-0.245**	-0.235**
		(9.92)	(-8.82)	(-8.83)	(-4.51)	(-2.16)	(-2.38)	(-2.05)
School fixed effects?	YES	YES	YES	YES	YES	YES	YES	YES
Time Fixed Effects?	YES	YES	YES	YES	YES	YES	YES	YES
School Individual Trends?	NO	NO	YES	YES	YES	YES	YES	YES
Observations	118758	117928	117928	117928	27292	27708	26419	27361

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.001

Note: Standard errors adjusted for clustering on the school level. All dependent variables are standardized to have a mean of zero and a standard deviation of unity within each year in which there are average grades of schools.

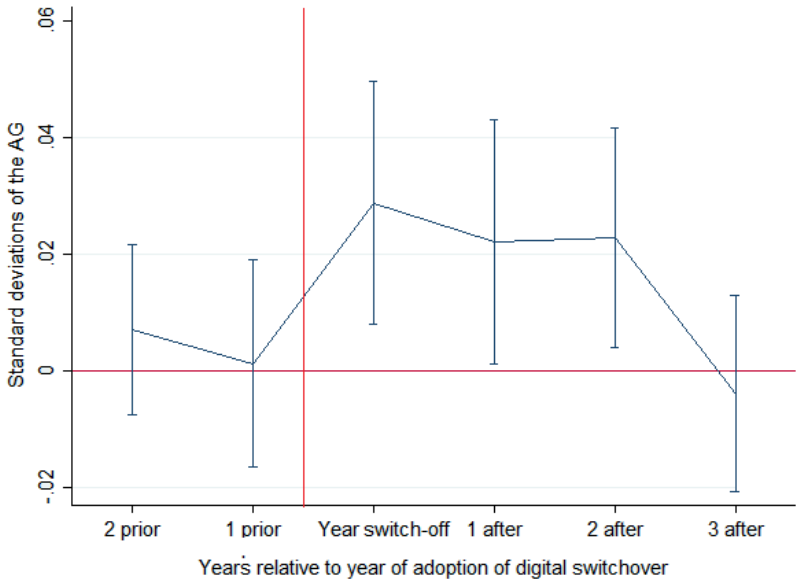

Figure 7: Estimated impact of digital switchover on average grades for years before, during, and after adoption

system occurred. These six indicator variables are equal to one in the relevant year and zero otherwise. More formally:

$$y_{s,t} = \alpha + \beta \sum_{r=-2}^{3} YearsinceSwitch_{s,t+r} + \gamma X_{s,t} + \delta_s + \lambda_t + \delta_s t + \varepsilon_{s,t} \qquad (4)$$

where I include school time varying characteristics, time effects, school fixed effects, and school individual trends. The estimated leads and lags of my benchmark specification are plotted in Figure 1. The coefficients of the leads are insignificant and close to zero, giving evidence that an anticipatory response of students is unlikely. Estimates of the year of the switchover and of the two years after it are positive and significant, indicating that average grades increase during the first three years since the switchover took place. However, the effect dilutes over time until I find an insignificant and close-to-zero estimate for the third year after the switchover, suggesting that it does not have long lasting effects. A possible explanation is that cohorts of individuals who have been just affected by the digital switchover may be enhancing their habits and the activities they do, improving their academic performance. However, after some time children get used to the availability of a higher number of channels and the effect dilutes.

4.5 Heterogeneity

My results have so far been centred on the effect that the digital switchover had on the average academic performance of schools. In many cases, however, it is relevant to study how this effect changes depending on the characteristics of schools, especially when looking at the sociodemographic attributes of their students. Prior literature shows a positive relationship between background socioeconomic status and school performance (Sirin (2005); White (1982)). Alternatively, television has been studied to improve language academic performance for the minorities that do not have English as a first language (Gentkow & Shapiro (2008)). If in-home learning is greater in households with more advantaged backgrounds or with higher educational resources, then, an increase in the supply of TV channels, which makes television more appealing, may be more beneficial for disadvantaged students, since it will displace activities and habits with different educational value depending on the socioeconomic status of students. Alternatively, it could happen that socioeconomic advantaged students choose to watch relatively more educational value added TV programs than disadvantaged students (Wright et al. (2001)). The introduction of new TV contents may change the type of programmes that children watch, the intrinsic values they obtain through watching TV, and also the way in which they behave or the activities they do, depending on their socioeconomic status. In column 1 of Table 4, I estimate specification (2) adding a term that is an interaction between the dummy that accounts for the digital switchover introduction and a dummy that takes value 1 if the school has more than 40% of students eligible for free school meals or looked after (25% of the sample satisfies this condition), and 0 otherwise . The estimated effect of digital switchover on average test scores is almost the triple for schools with a high proportion of students with free school meals, so that the digital television actually benefits more to students who cannot dispose of large educational resources. The effect disappears for the disadvantaged group after a year with accessibility to digital television. An additional year since the switchover is positive, albeit insignificant, for the rest of the sample.

The next column compare schools depending on their total number of students by controlling for an interaction between the dummy that accounts for the digital switchover introduction and a dummy that takes value 1 if the school is below the average total number of students (237) and 0 otherwise. The effect is four times higher in bigger schools, suggesting that rural areas were not necessarily the more affected ones by the digital switchover.

Columns 3 compares schools with a high and low proportion of students with English as first language. The estimated effect for non-English students is almost the triple than for native students, indicating that digital television improves the language and verbal abilities of the minorities that are not English native speakers. Still, the effect continues being positive and significant for the rest of the sample, suggesting that the improvement of language skills of non-natives was not the only mechanism through which the digital switchover improved the academic performance of students.

The following column test whether the digital switchover affected differently schools depending on their proportion of students with special needs. For this purpose, I compare schools with a proportion of students with special needs that is higher than 5.5% (10% of the sample), with the rest of schools. The estimates indicate that the effect is negative for the group of schools with a high proportion of students with SEN, and positive for the rest of schools, so that the availability of a higher number of channels does not seem to benefit students with SEN.

Finally, I compare religious with non-religious schools in the last column of Table 4, finding similar estimates for both categories, even though the effect seems to dilute more quickly in non-religious schools.

The evidence that the students who benefit more from the accessibility to digital television are precisely those who are more disadvantaged either in socioeconomic or language terms, confirms the results presented in Table 3, where the positive impact that digital television had on the cognitive development was driven by students at the bottom quartiles of the grades distribution. Given that disadvantaged or poor performance students are the ones who are more likely to be involved in low educational value activities, the results support the hypothesis that the impact of television exposure on education depends critically on the educational value of the alternative activities that it crowds out (Gentkow and Shapiro, 2008). The following sections corroborate whether the digital switchover actually reduced low educational value added activities , and give an explanation of the mechanism through which this could have happened.

4.6 Robustness checks

I next test the robustness of the results by (i) using different dependent variables, (ii) controlling for a richer set of covariates including demographic covariates, LEA individual trends and LEA-by-year effects (iii) running specification (2) in subsamples including data on a different number of years, and (iv) using weighted regressions on the number of total students. Estimates are found in Table 6. A first specification addresses the concern of whether the improvement in the academic performance is coming from an enhancement in the verbal or maths abilities. Columns 1 and 2 of Table 5 re-estimate my benchmark specification (2) using the percentage of students obtaining a grade of 25 or higher in the Maths and in the English part as a dependent variable, respectively. The estimated impact is positive and significant in both sections of the exam, but higher in Maths one. In particular, digital switchover increases by 0.37 and 0.26 percentage points the probability of a student obtaining a score of 25 or higher in Maths and English respectively. The effect of an additional year since analogue switch-off is negative, so that the effect dilutes over time in both sections, even though faster in the English part of the exam. Column 3 uses the CVA as dependent variable, to add robustness to my analysis. The CVA measure is a statistical mean of assessing the relative effectiveness of a school or measuring pupil progress, and it is calculated by the UK Department

Table 4: Heterogeneity

	Disadvantaged	Size school	English 1st language	SEN	Religious
Digital Switchover	0.0234**	0.0535***	0.0314**	0.0398***	0.0373**
	(1.99)	(4.36)	(2.79)	(3.59)	(3.01)
Digital TV tenure	0.00499	-0.0236**	-0.00207	-0.0132	-0.0169
	(0.42)	(-1.98)	(-0.18)	(-1.13)	(-1.40)
N students with SEN	-0.0644***	-0.0645***	-0.0642***	-0.0639***	-0.0641***
	(-56.46)	(-56.37)	(-56.43)	(-55.96)	(-55.81)
lnTOTPUPS	-0.407***	-0.428***	-0.438***	-0.431***	-0.413***
	(-8.41)	(-8.87)	(-9.06)	(-8.92)	(-8.47)
Digital Switchover × disadvantaged	0.0502**				
	(2.70)				
Digital TV tenure × disadvantaged	-0.0612***				
	(-6.87)				
Digital Switchover × sizeschool		-0.0373**			
		(-2.45)			
Digital TV tenure × sizeschool		0.0310***			
		(4.38)			
Digital Switchover × englishlanguage			0.0501*		
			(1.93)		
Digital TV tenure × englishlanguage			-0.0701***		
			(-5.70)		
Digital Switchover × SENdummy				-0.0682**	
				(-2.19)	
Digital TV tenure × SENdummy				0.0371**	
				(2.63)	
Digital Switchover × religious					-0.00978
					(-0.61)
Digital TV tenure × religious					0.0195**
					(2.64)
School Effects?	YES	YES	YES	YES	YES
Time Effects?	YES	YES	YES	YES	YES
Individual school trends?	YES	YES	YES	YES	YES
Observations	117928	117928	117928	117928	116772

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.001

Note: Standard errors adjusted for clustering on the school level. All dependent variables are standardized to have a mean of zero and a standard deviation of unity within each year in which there are average grades of schools.

of Education by means of a statistical model based on the actual test results of the given year group. This model calculates the national average results attained by each category of pupil the statistical prediction and compares each individuals test/exam results against that prediction. I find similar estimates than my baseline results.

Columns 3-5 check whether the impact that digital switchover has on grades change as a consequence of using subsamples with a different number of years in the analysis. In particular, I use periods 2006-2015, 2008-2014 and 2009-2013 respectively in columns 5, 6 and 7, to test how the impact on grades changes as I narrow the window of time around the switchover date. The estimates of the effect of the digital switchover on test scores are very stable across specifications. In contrast, the effect of an additional year since the switchover gets smaller and insignificant as the window of time is narrowed. A possible explanation is that, as the period of time that I use for the analysis is shortened, the number of years in which the digital television has been in place since the switchover are also reduced. In columns 69, I add controls for quadratic school time trends, LEA individual trends, and interactions between year dummies and indicators for each of the LEAs, allowing standardized average grades to trend nonlinearly and also absorbing region specific shocks. The implied digital switchover coefficient is largely insensitive to these specifications. Finally, in column 10, I make use of weighted regressions on the number of students in a school, which does not alter my results.

4.7 Impact of Digital Television on Habits

This paper studies the impact of the digital switchover on the cognitive development of children. I find that the availability to a higher number of TV channels enhanced the academic performance of students at the last stage of primary education by 0.028 standard deviations, but that the effect dilutes over time.

In this section, I evaluate the hypothesis of whether this positive impact comes as a response of students reducing the time they dedicate to low educational value added or even harmful activities once they have access to digital television. For this purpose, I use the Understanding Society Survey data, merged with data on the date of the digital switchover at the postcode level. I then exploit the timing of the switch off of the analogue system, to use a difference-in-difference design that compares the information provided by interviewed children who were affected by the digital switchover at different dates, and before and after the switch to digital signal occurred. More formally, I estimate the following specification:

$$a_{i,t} = \alpha + \beta DigitalSwitch_{s,t} + \phi DigitalTenure_{s,t} + \gamma X_{i,t} + \delta_i + \lambda_t + \varepsilon_{i,t} \quad (5)$$

Where δ_i are child fixed effects and λ_t are time year fixed effects. I do not control for individual child trends because I do not have enough number

Table 5: Robustness checks

	% students above 25 in Maths	% students above 25 in English	Contextual Value Added	Standarized KS2 Average Grade	Standarized KS2 Average Grade	Using different samples: Standarized AG 2006-2015	Using different samples: Standarized AG 2008-2014	Using different samples: Standarized AG 2009-2013	Standarized KS2 Average Grade
Digital Switchover	0.00322**	0.00265*	0.0719***	0.0356***	0.0355*	0.0287**	0.0188	0.0309*	0.0417***
	(2.01)	(1.73)	(4.90)	(3.60)	(1.90)	(2.71)	(1.53)	(1.91)	(3.91)
Digital TV tenure	-0.00206	-0.00365**	-0.0402**	-0.0144	-0.0242*	-0.00281	0.0140	-0.00203	-0.0172
	(-1.26)	(-2.36)	(-2.55)	(-1.61)	(-1.88)	(-0.26)	(0.95)	(-0.09)	(-1.50)
N students with SEN	-0.00764***	-0.00797***	-0.0178***	-0.0629***	-0.0639***	-0.0636***	-0.0735***	-0.0746***	-0.0500***
	(-47.57)	(-52.55)	(-13.72)	(-60.00)	(-60.60)	(-59.34)	(-55.88)	(-44.15)	(-42.67)
lnTOTPUPS	-0.0646***	-0.0343***	-0.821***	0.188***	0.186***	-0.393***	-0.455***	-0.407***	-0.407***
	(-8.99)	(-5.47)	(-12.94)	(6.12)	(6.02)	(-8.87)	(-7.13)	(-4.24)	(-8.47)
School FE?	YES	YES	YES	YES	YES	YES	YES	YES	YES
Time FE?	YES	YES	YES	YES	YES	YES	YES	YES	YES
School individual Trends?	YES	YES	YES	NO	NO	YES	YES	YES	YES
LEA individual trends?	NO	NO	NO	YES	NO	NO	NO	NO	NO
LEA and year specific shocks?	NO	NO	NO	NO	YES	NO	NO	NO	NO
Weighted regression?	NO	NO	NO	NO	NO	NO	NO	NO	YES
Observations	118108	118051	117485	117928	117911	130487	91722	65003	117928

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.001

Note: Standard errors adjusted for clustering on the school level. All dependent variables are standardized to have a mean of zero and a standard deviation of unity within each year in which there are average grades of schools.

of periods to do so for many of the respondents. I am able to distinguish when each postcode area was affected by the digital switchover. Let $DigitalSwitch_{s,t}$ be a dummy that takes value 1 if the postcode area s, where child i lives, has already been affected at year t by the digital switchover, and 0 otherwise. The dependent variable $a_{i,t}$ accounts for the frequency in which child i do an activity at time t, or the probability of doing it. More specifically, I use specification (4) to study the effect of the digital switchover on five leisure activities: whether the child has ever drunk alcohol, smoked, and the frequency in which he has drunk alcohol, smoked or arrived late at home during the last week. $X_{i,t}$ is a set of demographic characteristics for each child answering to the survey, which change across children and over time. In particular, $X_{i,t}$ includes the sex and age of child i; the total number of individuals, children aged between 10 and 14, and adults older than 14 who live in the household of individual i; the gross income, and the number of cars and bedrooms at home that the household of child i has; whether the child lives in a rural area, and whether the parents of the child needed a translation of the interview, and therefore are not English native speakers.

In table 6, I estimate specification (4) using as dependent variables the frequency in which a child arrives later than 9 at home, whether the child has tried alcohol or smoked, and the frequency in which the child drinks or smokes. The frequency of activities are measured by a categorical variables that take 5 possible values, with a higher value indicating that the harmful activity is done more frequently. The variables of having ever tried alcohol or smoked are dummy variables, which take value 0 when the child has never done it, and 1 otherwise. Importantly, digital switchover reduces the frequency in which a child aged from 10 to 15 drinks alcohol, and also reduces the probability of having drunk at some point in life by 5.2 percentage points. The reductions in the frequency of drinking and in the probability of having tried alcohol dilutes over time, given that the coefficient of an additional year of digital television is positive, albeit insignificant in the case of the frequency of drinking. Digital switchover also decreases the frequency of smoking, the probability of having smoked, and the frequency in which the child arrives late home, albeit these estimates are insignificant. The effects also seem to dilute over time, except in the case of the frequency in which the child smokes, which decreases even more over time. These results suggest that the positive impact that the availability to a higher number of channels had on the cognitive development of children comes through a reduction in some harmful activities that children do in their leisure time, as a response of the digital switchover.

The impact that the digital switchover had on the academic performance and habits of children may have occurred via two different mechanisms, the quantity and the quality mechanism. The former lies in the digital switchover having induced children to watch more TV instead of doing lower educational value added activities, and the latter consists on new TV contents having benefited behaviour and habits through the intrinsic values that new TV programmes transmit, and so improving motivation and cognitive development.

Table 6: Harmful activities and bad habits

	Times arriving late at night	Has ever smoked	Frequency in which smokes	Frequency in which drinks alcohol	Has ever drunk alcohol
Digital Switchover	-0.00545	-0.0105	-0.00248	-0.0731**	-0.0522***
	(-0.27)	(-1.19)	(-0.13)	(2.32)	(-3.50)
Digital Tenure	0.0108	0.000469	-0.00855	0.00336	0.0315***
	(0.98)	(0.10)	(-0.83)	(-0.19)	(3.81)
Female	0.319	0.0167	0.0243	-1.865***	0.639**
	(1.01)	(0.12)	(0.08)	(-3.56)	(2.70)
N people in hh	0.0867	-0.167	-0.190	0.0434	-0.0136
	(0.36)	(-1.53)	(-0.83)	(0.13)	(-1.15)
Hh gross income	0.00413	-0.000196	0.000919	0.000974	-0.00306
	(1.18)	(-0.12)	(0.28)	(0.17)	(-1.16)
Age	-0.0377	0.0130	0.0535*	-0.00365	0.00230
	(-1.27)	(0.99)	(1.93)	(-0.08)	(0.10)
Rural area	-0.185**	0.0146	-0.0459	0.169	0.0644
	(-2.39)	(0.42)	(-0.63)	(1.24)	(1.11)
N bedrooms in hh	-0.00222	-0.00780	-0.0210	0.0742**	-0.0206
	(-0.11)	(-0.88)	(-1.13)	(2.34)	(-1.39)
N cars in hh	-0.0282**	0.00567	-0.00198	-0.0197	0.00157
	(-2.00)	(0.90)	(-0.15)	(-0.90)	(0.15)
Non English native	0.0447	0.00656	0.00560	0.00543	0.0183
	(1.37)	(0.45)	(0.18)	(0.10)	(0.74)
N children in hh	-0.102	0.130	0.124	0.0596	-0.0197**
	(-0.42)	(1.19)	(0.55)	(0.18)	(-2.04)
N adults in hh	-0.0568	0.161	0.197	-0.0520	
	(-0.23)	(1.48)	(0.87)	(-0.16)	
Children FE?	YES	YES	YES	YES	YES
Time FE?	YES	YES	YES	YES	YES
Constant	1.223**	-0.0818	-0.371	7.018***	-0.793*
	(2.08)	(-0.31)	(-0.68)	(7.24)	(-1.80)
Observations	14867	14842	14819	11505	14812

t statistics in parentheses
* p<0.10, ** p<0.05, *** p<0.001

To test whether the quantity mechanism drove the increase in academic performance and reduction in harmful habits, I use two categorical variables that take values from 1 to 5, to respectively measure the time that child i dedicates to watch television during the week days and weekend, and include them as dependent variables in specification (4). I find that estimates do not follow the same direction and are insignificant, with the exception of the positive impact of an additional year of digital television availability on the TV watching time during weekends. The improvement in grades and habits seems not to be driven by the quantity mechanism.

To test for the quality mechanism, I implement my baseline specification (4), using as dependent variable the answer to several questions about intrinsic values in the youth questionnaire, which are measured by categorical variables that take a higher number for higher values of the intrinsic value which is being measured. I group the variables that I use to measure the quality mechanism in three main groups: (i) motivation to obtain more education, (ii) interest, concentration and attention in school, and (iii) culture and concerns with social problems. New TV contents may have enhanced behaviour and habits by increasing any of the previous groups of intrinsic values.

Estimates are presented in Table 7 and 8. As regard to motivation, I find that digital switchover has a positive and significant impact on the happiness towards school work and on the inclination to take tertiary education, suggesting that it increased motivation of children towards education. Digital switchover does not have a significant impact on any other measure of motivation, but an additional year since the switchover decreases the confidence that children have on themselves, contributing to the dilution of the effect that digital switchover has on grades and habits. On the side of attentional problems, I find that digital switchover increases confidence, decreases concentration problems and the frequency in which a child misbehaves at school, albeit insignificant. I also find that it has a negative and insignificant impact in constancy. The balance of accessibility to a higher number of channels on attentional problems seems to be positive, although insignificant. Finally, there is mixed evidence of the impact that the digital switchover had on social concerns. While the digital switchover increases the frequency in which children volunteer, it also leads children to care less about the others, be less helpful, and less interested in politics. These estimates are insignificant.

This suggests that the impact that digital television had on academic performance and habits was generated by an increase in the motivation, and so, through the quality mechanism. More generally, new TV contents increased motivation of children to obtain education, which improved the way they behave. It is important to bear in mind three possible limitations. First of all, the quantity and quality mechanisms are imperfectly measured by using a set of questions as a proxy. Secondly, the answers to the questions in the Understanding Society survey are self-completed, so that the response to them may be subjectively biased. Finally, some of the questions that measure my quantity and quality channel are only present in some of the years in which the Understanding Society survey was carried out, which reduces the precision of some

of my estimates. The last limitation does not play a role in my measure of happiness with school work however, given that this question is present in every year in which the survey was carried out. In summary, even though the last set of regressions suggests that the quality mechanism, and in particular motivation, was the cause of the positive impact that the digital switchover had on grades and harmful habits, I cannot disregard the possibility that the quantity mechanism, or that the quality mechanism via an improvement in attention or social concerns, may have also been the mechanism through which the cognitive development and behaviour were improved.

5 Conclusions

This paper studies the effect of digital television on cognitive development and habits. For this purpose, I use the digital switchover that took place from 2008 to 2012 in UK. This induces a natural experiment: the timing of entry of the digital switchover by stages in the different postcode areas. I firstly estimate whether a sudden increase in the number of TV channels, as part of the analogue switch-off, has an effect on the academic performance of children at their last stage of primary education. I then test whether the impact on grades comes as a response of a change in the habits of children, and lately, I study whether the effect on grades and habits is caused by a quantity or a quality mechanism.

I find a significand and positive effect of the digital switchover on academic performance. Schools in postcode areas where the digital switchover has already occurred score 0.028 standard deviations higher on KS2 average scores than schools that had still not been affected by the analogue switch-off, suggesting that digital switchover contributed to the academic development and formation of human capital. The effect was driven by educational institutions performing at the bottom of the test scores distribution before the digital switchover started, with the lowest quartile of schools improving the average grade by 0.071 standard deviations, and with the top quartiles having no significant boost in marks, which suggests that digital switchover reduced the educational inequality in England. Importantly, the positive impact is a short-term effect, disappearing three years after digital switchover has been implemented. I also find that the analogue switch-off reduces both the probability of doing and the frequency in which students do some harmful habits. In particular, the digital switchover decreases the probability of a child having ever drunk alcohol by 7.31 percentage points, and also the frequency in which children drink alcohol. The accessibility to digital television also decreases other harmful activities such as smoking or arriving late at home, albeit insignificant. These results are consistent with the effect that I find on academic achievement, and confirm the hypothesis that television exposure impact on education depends critically on the educational value of the alternative activities that it crowds out (Gentkow and Shapiro, 2008). Two possible explanations of the improvement

Table 7: Quantity or quality channel

	Quantity channel		Quality channel: Motivation						
	TV freq during school days	TV freq during weekend	Happiness with school work	GCSE importance	Want to go to university	Time doing homework week days	Time doing homework weekend	Feel useless	Solve own problems
Digital Switchover	-0.0311	0.0122	-0.0841**	0.00150	0.0158*	-0.0744	-0.0686	0.0556	0.0207
	(-1.25)	(0.21)	(-2.06)	(0.08)	(1.67)	(-0.31)	(-0.40)	(0.96)	(0.45)
Digital Tenure	0.00977	0.0707**	0.00330	0.00710	-0.00369	-0.119	-0.0646	-0.0226	-0.0622**
	(0.71)	(2.23)	(0.15)	(0.71)	(-0.71)	(-0.92)	(-0.70)	(-0.72)	(-2.47)
Female	1.157**	0.496	0.512	0.0415	-0.0000390	0.771	-0.532	0.0187	0.127
	(2.92)	(0.74)	(0.79)	(0.15)	(-0.00)	(0.29)	(-0.20)	(0.03)	(0.24)
Age	-0.0549	-0.0956	0.0472	-0.0133	-0.00523	-0.0656	-0.285	0.0700	0.0453
	(-1.48)	(-0.98)	(0.78)	(-0.49)	(-0.37)	(-0.17)	(-1.01)	(0.73)	(0.58)
Hh income	0.00694	-0.0110	-0.0110	-0.000935	0.000449	-0.0650	-0.0124	-0.000434	-0.000260
	(1.58)	(-0.87)	(-1.53)	(-0.29)	(0.28)	(-1.28)	(-0.34)	(-0.03)	(-0.03)
Rural area	0.0709	0.0671	0.102	-0.0960	0.0385	-0.276	0.613	0.107	0.287
	(0.71)	(0.27)	(0.61)	(-1.30)	(1.00)	(-0.27)	(0.81)	(0.43)	(1.43)
Not English native	0.0431	0.443***	0.0000987	-0.0203	0.0186	0.0418	0.324	0.209*	0.122
	(1.06)	(3.55)	(0.00)	(-0.68)	(1.17)	(0.08)	(0.90)	(1.73)	(1.24)
N bedrooms in hh	0.0339	0.0757	-0.0626	-0.0182	0.00378	0.400	0.215	-0.175**	-0.00806
	(1.38)	(1.16)	(-1.54)	(-1.02)	(0.41)	(1.51)	(1.13)	(-2.71)	(-0.15)
N cars in hh	0.0162	0.0729	0.0188	0.00607	-0.00838	0.270	0.0166	-0.0133	0.0294
	(0.92)	(1.58)	(0.65)	(0.47)	(-1.21)	(1.43)	(0.12)	(-0.29)	(0.80)
N people in hh	0.0436	0.0849	1.380**	-0.289	0.00634	0.0344	0.108	-0.0447	-0.00800
	(0.14)	(1.61)	(2.76)	(-1.32)	(0.07)	(0.16)	(0.69)	(-0.86)	(-0.19)
N children in hh	-0.0113	0.0625	-1.420**	0.224	0.000994	-0.0644	0.0730	-0.0344	0.0382
	(-0.04)	(1.42)	(-2.84)	(1.03)	(0.01)	(-0.36)	(0.57)	(-0.79)	(1.09)
N adults in hh	-0.0122	0	-1.410**	0.283	-0.00935	0	0	0	0
	(-0.04)	(.)	(-2.82)	(1.30)	(-0.10)	(.)	(.)	(.)	(.)
Children FE?	YES	YES	YES	YES	YES	YES	YES	YES	YES
Time FE?	YES	YES	YES	YES	YES	YES	YES	YES	YES
Constant	1.291*	2.128	1.560	1.758***	0.902**	0.187	3.124	-3.274**	-3.281**
	(1.75)	(1.34)	(1.29)	(3.31)	(2.97)	(0.03)	(0.57)	(-2.10)	(-2.59)
Observations	14900	5916	14849	14532	10802	5699	5673	5898	5898

t statistics in parentheses
* p<0.10, ** p<0.05, *** p<0.001

Table 8: Quantity or quality channel

	Quality channel: Attention				Quality channel: Social Concerns				
	Difficult to concentrate	Loses confidence easily	Finishes started work	Misbehaves often at school	Cares about others feelings	Helpful if someone is hurt	Often volunteers to help	Frequency discussing TV at home	Degree of interest in politics
Digital Switchover	-0.0151	-0.0481	-0.000917	-0.0732	-0.00106	-0.0290	0.0171	-0.0641	-0.0225
	(-0.38)	(-1.19)	(-0.03)	(-1.56)	(-0.04)	(-0.93)	(0.47)	(-1.04)	(-0.60)
Digital Tenure	0.00553	-0.00339	-0.00923	0.0338	0.000929	-0.00309	-0.00443	0.0195	-0.00259
	(0.30)	(-0.18)	(-0.60)	(1.57)	(0.07)	(-0.21)	(-0.26)	(0.59)	(-0.15)
Female	0.903	0.0770	-1.137	-0.957	-0.966*	0.0308	-0.144	0.387	0
	(1.08)	(0.09)	(-1.62)	(-0.98)	(-1.67)	(0.05)	(-0.19)	(0.55)	(.)
Age	-0.0181	0.0756	-0.0101	0.0947	0.0409	-0.105**	-0.0188	-0.0388	-0.101
	(-0.26)	(1.10)	(-0.18)	(1.18)	(0.87)	(-1.97)	(-0.30)	(-0.38)	(-1.46)
Hh income	-0.00000371	0.00000533	-0.00000163	0.0000129	-0.00000532	-0.00000203	-0.00000118	0.0000161	-0.00000960
	(-0.52)	(0.74)	(-0.27)	(1.52)	(-1.09)	(-0.36)	(-0.18)	(1.20)	(-1.23)
Rural area	-0.222	-0.0689	0.0608	0.0814	0.0801	-0.102	-0.249*	0.353	-0.168
	(-1.38)	(-0.43)	(0.45)	(0.45)	(0.74)	(-0.83)	(-1.73)	(1.34)	(-1.09)
Not English native	0.0856	0.0250	-0.0356	-0.0299	0.0526	-0.00342	-0.00477	0.172	-0.137*
	(1.32)	(0.38)	(-0.65)	(-0.39)	(1.18)	(-0.07)	(-0.08)	(1.30)	(-1.95)
N bedrooms in hh	-0.00434	-0.0554	0.0245	0.0103	0.0191	-0.0212	0.0402	-0.0803	-0.0189
	(-0.11)	(-1.45)	(0.78)	(0.24)	(0.73)	(-0.72)	(1.17)	(-1.15)	(-0.38)
N cars in hh	0.0463*	-0.0287	-0.00876	-0.0116	0.0102	0.00738	0.00563	-0.0518	-0.00611
	(1.65)	(-1.01)	(-0.37)	(-0.35)	(0.53)	(0.34)	(0.22)	(-1.07)	(-0.21)
N people in hh	0.431	0.209	0.371	0.0380	-0.00909	0.0127	-0.0500	0.0315	0.0459
	(1.15)	(0.55)	(1.18)	(0.09)	(-0.04)	(0.04)	(-0.15)	(0.56)	(0.16)
N children in hh	-0.431	-0.185	-0.430	0.0272	0.0379	-0.0285	0.0127	-0.00540	-0.00493
	(-1.15)	(-0.49)	(-1.37)	(0.06)	(0.15)	(-0.10)	(0.04)	(-0.12)	(-0.02)
N adults in hh	-0.413	-0.212	-0.419	-0.0641	0.0427	0.00123	0.0972	0	-0.00332
	(-1.10)	(-0.56)	(-1.34)	(-0.15)	(0.17)	(0.00)	(0.29)	(.)	(-0.01)
Children FE?	YES	YES	YES	YES	YES	YES	YES	YES	YES
Time FE?	YES	YES	YES	YES	YES	YES	YES	YES	YES
Constant	0.746	1.065	4.424***	-4.002**	3.266**	3.907***	2.960**	1.343	4.262***
	(0.50)	(0.71)	(3.56)	(-2.31)	(3.20)	(3.38)	(2.20)	(0.81)	(4.92)
Observations	8814	8810	8814	8814	8843	8819	8812	5902	5189

t statistics in parentheses
* p<0.10, ** p<0.05, *** p<0.001

in grades and habits are: either accessibility to digital television increased TV time viewing displacing harmful activities, and so academic performance of the more problematic children, or it increased intrinsic values such as motivation or consciousness about the negative implications of having bad habits, which enhanced habits and academic performance. I test for these two mechanisms in the final part of the paper, finding that the positive impact of the digital switchover on grades and habits comes through the quality mechanism, where motivation to study is increased as a response of the new TV contents. Despite finding that the digital switchover increases motivation, I am not able to reject the possibility that the quantity mechanism, and so a change in the time allocation, was one of the reasons driving the enhancement in grades and habits.

This paper leaves a number of open questions on the impact of the digital switchover. First, the paper makes use of a school level dataset to analyse the effect of digital TV accessibility on school achievement. While it is true that the analysis could benefit from using a student level dataset, it is also true that KS2 tests are taken every year by a different cohort of students, so that it would not be possible to account for unobserved student fixed effects even if I used a pupil level dataset on the KS2 performance. Another possibility is to compare the difference in the performance between the KS2 and the KS4 evaluation, which is the test that follows the KS2 exam, across students who have been affected and who have not by the digital switchover in the meantime between both assessments. In ongoing research, I use the national pupil dataset in order to study the effect of the digital switchover using individual data. However, it is important to bear in mind that, once students have taken the KS2 test, they usually take five years to do KS4 tests, meaning that the analysis may be subject to an important amount of unobserved heterogeneity that could have occurred in the meantime between both examinations. Second, I measure my quantity and quality mechanism using a set of self-completed questions about emotions, habits, and intrinsic values as a proxy, which are imperfect measures of both mechanisms and are subject to personal biases. Even though I am not able to use alternative private data given budget constraints, the use of non-public datasets that give more specific information about the TV time viewing of children living in different postcode areas in England, may help to test my hypothesis of whether the impact on cognitive development and habits is being driven by the quantity or quality mechanism. Finally, I hope that more evidence on the effect of other sources of technology shocks, such as the introduction of internet broadband connection, or the implementation of mobile data and smartphones, will complement the evidence in this paper.

Level	Point Score	Description
3a	23	Below Average
4c	25	About Average
4b	27	About Average
4a	29	Above National Average
5c	31	Above National Average
5b	33	Well Above National Average
5a	35	Well Above National Average
6c	37	Exceptional
6b	39	Exceptional

Appendices

A Mapping of point scores to levels

B Switchover timetable

TV region	Transmitter group	Area served	Homes (000s)	Stage One	Stage Two
Border	Selkirk	Scottish Borders	52	6 Nov 08	20 Nov 08
	Douglas	the Isle of Man	38	18 Jun 09	16 Jul 09
	Caldbeck	Cumbria, south west Scotland and the south Lakes	308	24 Jun 09	22 Jul 09
West Country	Beacon Hill	Torbay and south Devon	132	8 Apr 09	22 Apr 09
	Stockland Hill	Exeter, parts of Devon, Somerset and Dorset	206	6 May 09	20 May 09
	Huntshaw Cross	north Devon	66	1 Jul 09	29 Jul 09
	Redruth	west Cornwall and the Isles of Scilly	122	8 Jul 09	5 Aug 09
	Caradon Hill	Plymouth, parts of Devon and east Cornwall	255	12 Aug 09	9 Sep 09
Wales	Kilvey Hill	the Swansea area	132	12 Aug 09	9 Sep 09
	Preseli	much of south west Wales	83	19 Aug 09	16 Sep 09
	Carmel	parts of south and central Wales	110	26 Aug 09	23 Sep 09
	Llanddona	north west Wales	95	21 Oct 09	18 Nov 09
	Moel y Parc	north east Wales	179	28 Oct 09	25 Nov 09
	Long Mountain	parts of east and central Wales	24	4 Nov 09	3 Dec 09
	Blaenplwyf	parts of west and central Wales	26	10 Feb 10	10 Mar 10
	Wenvoe	Cardiff, Newport and south east Wales	649	3 Mar 10	31 Mar 10
Granada	Winter Hill	Liverpool, Manchester, Lancashire, Cheshire and north Staffordshire	3,041	4 Nov 09	2 Dec 09
West	Mendip	Bristol, parts of Somerset, Dorset, Wiltshire and Gloucestershire	913	24 Mar 10	7 Apr 10
STV North	Bressay	the Shetland Islands	9	5 May 10	19 May 10
	Keelylang Hill	the Orkney Islands	10	12 May 10	26 May 10
	Rumster Forest	Caithness and North Sutherland	23	2 Jun 10	16 Jun 10
	Eitshal	Lewis, Wester Ross, north west Sutherland and parts of Harris and Skye	12	7 Jul 10	21 Jul 10
	Skriaig	Skye, Harris, North Uist, Benbecula, South Uist and parts of Barra	8	14 Jul 10	28 Jul 10
	Angus	Angus, Dundee, Perth and parts of Fife	210	4 Aug 10	18 Aug 10
	Durris	Aberdeen and Aberdeenshire	221	1 Sep 10	15 Sep 10
	Knockmore	Morayshire, Strathspey and parts of Easter Ross	39	8 Sep 10	22 Sep 10
	Rosemarkie	Inverness and the Great Glen	70	6 Oct 10	20 Oct 10
Channel	Fremont Point	the Channel Islands	50	17 Nov 10	N/A
STV Central	Torosay	south west Highlands and Islands	17	13 Oct 10	27 Oct 10
	Darvel	parts of central Scotland, Argyll and Bute	197	11 May 11	25 May 11
	Rosneath	Rosneath	44	11 May 11	25 May 11
	Craigkelly	Lothian, parts of Edinburgh and parts of Fife	455	1 Jun 11	15 Jun 11
	Black Hill	Glasgow, central Scotland and parts of Edinburgh	1,020	8 Jun 11	22 Jun 11

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TV region	Transmitter group	Area served	Homes (000s)	Stage One	Stage Two
Central	Nottingham	the Nottingham area	74	30 Mar 11	13 Apr 11
	Lark Stoke	the Stratford-upon-Avon area	38	6 Apr 11	20 Apr 11
	Bromsgrove	the Bromsgrove area	30	6 Apr 11	20 Apr 11
	Ridge Hill	Herefordshire, south Shropshire and parts of Gloucestershire	286	6 Apr 11	20 Apr 11
	The Wrekin	north Shropshire and south Cheshire	280	6 Apr 11	20 Apr 11
	Waltham	much of the East Midlands	815	17 Aug 11	31 Aug 11
	Sutton Coldfield	much of the West Midlands	2,123	7 Sep 11	21 Sep 11
	Fenton	Stoke-on-Trent and Newcastle-under-Lyme	128	7 Sep 11	21 Sep 11
	Oxford	Oxfordshire, parts of Berkshire and Buckinghamshire	415	14 Sep 11	28 Sep 11
Yorkshire	Oliver's Mount	the Scarborough area	31	3 Aug 11	17 Aug 11
	Belmont	Lincolnshire and East Yorkshire	712	3 Aug 11	17 Aug 11
	Sheffield	the Sheffield area	119	10 Aug 11	24 Aug 11
	Chesterfield	the Chesterfield area	34	10 Aug 11	24 Aug 11
	Emley Moor	most of Yorkshire	1,756	7 Sep 11	21 Sep 11
Anglia	Sandy Heath	Cambridgeshire, Northamptonshire and Bedfordshire	966	30 Mar 11	13 Apr 11
	Sudbury	parts of Suffolk and Essex	523	6 Jul 11	20 Jul 11
	Tacolneston	Norfolk and north Suffolk	408	9 Nov 11	23 Nov 11
Meridian	Hannington	parts of Hampshire, Berkshire and Surrey	481	8 Feb 12	22 Feb 12
	Midhurst	much of West Sussex	99	29 Feb 12	14 Mar 12
	Whitehawk Hill	the Brighton area	119	7 Mar 12	21 Mar 12
	Rowridge	Hampshire, the Isle of Wight, parts of Dorset, Wiltshire and West Sussex	728	7 Mar 12	21 Mar 12
	Tunbridge Wells	the Tunbridge Wells area	54	30 May 12	13 Jun 12
	Heathfield	East Sussex and south west Kent	214	30 May 12	13 Jun 12
	Hastings	the Hastings area	20	30 May 12	13 Jun 12
	Bluebell Hill	north and mid Kent	217	13 Jun 12	27 Jun 12
	Dover	south and east Kent, and parts of East Sussex	238	13 Jun 12	27 Jun 12
London	Crystal Palace	Greater London and parts of the Home Counties	4,858	4 April 12	18 April 12
Tyne Tees	Bilsdale	Teesside and much of North Yorkshire	595	12 Sep 12	26 Sep 12
	Chatton	north Northumberland	33	12 Sep 12	26 Sep 12
	Pontop Pike	Tyne and Wear and County Durham	790	12 Sep 12	26 Sep 12
Northern Ireland	Limavady	north west Northern Ireland	104	10 Oct 12	24 Oct 12
	Brougher Mountain	south west Northern Ireland	37	10 Oct 12	24 Oct 12
	Divis	Belfast and the surrounding area	508	10 Oct 12	24 Oct 12

[H]

Figure 9: Switchover timetable

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